Growth Pulse
SMART PLANT MONITORING

Technical Reference Manual

The current-state specification: hardware, firmware, cloud, schemas, and APIs

Document	GP-TR-001
Revision	1.0
Date	June 11, 2026
Status	Released
Applies to	Firmware v4.3 / Web app v2.22.2

Contents

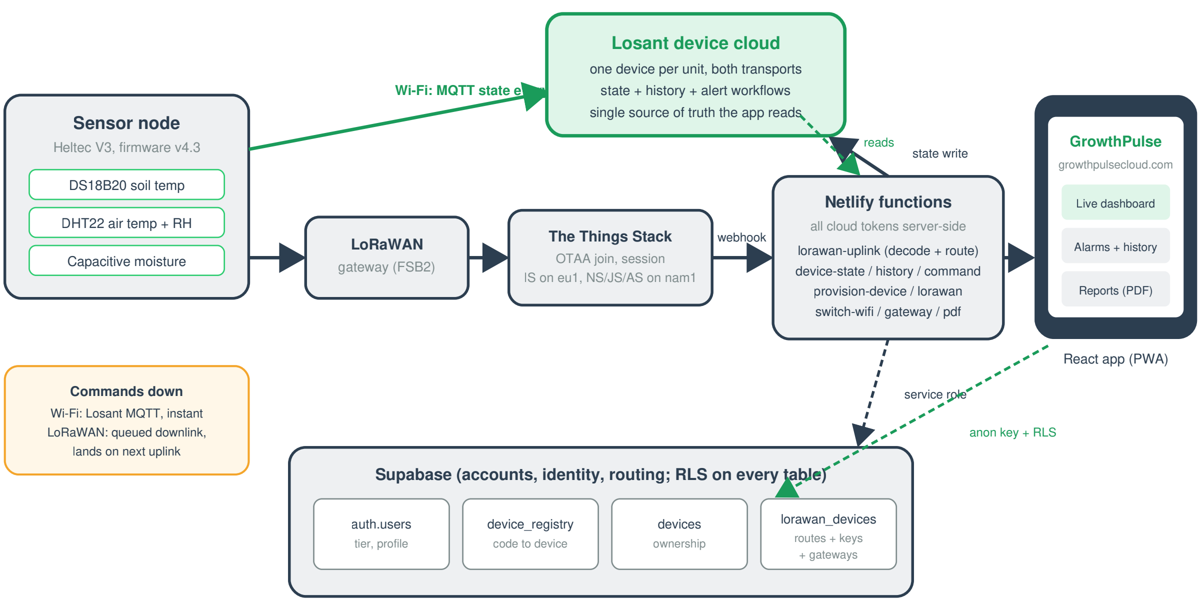
About this manual

This is the current-state specification of GrowthPulse: every pin, constant, identifier, payload, endpoint, schema, and configuration value in the shipping system, in reference form. It states what the system is; the Engineering Manual explains why it is that way, and the Operations Manual explains how to work on it. The source code is authoritative; if this document and the code disagree, correct this document.

1. System versions and identifiers

Item	Value
Production firmware	GP_Combined v4.3
Web application	v2.22.2
Live application URL	https://growthpulsecloud.com
Source repository	github.com/BryanPuckettGH/growthpulse-app
Losant application device class	standalone, one device per physical unit
TTS application id	growthpulse
TTS deployment	The Things Stack Sandbox (The Things Network)
LoRaWAN region	US915, sub-band 2 (FSB2)
Demo unit pairing code / hotspot	4A7AC / GrowthPulse-04A7AC
Demo unit Losant device id	6a1fb486df527c8bf8d3324b
Development gateway	Elecrow ThinkNode G1, EUI E4:38:19:FF:FE:2A:58:80

GrowthPulse System Architecture



Both transports write to the same Losant device, so a node stays one plant in the app no matter how it connects.

2. Hardware reference

2.1 Board

Heltec WiFi LoRa 32 V3 (silkscreen HTIT-WB32LAF): ESP32-S3 dual-core at 240 MHz, 8 MB flash, 2.4 GHz Wi-Fi b/g/n, BLE, Semtech SX1262 sub-GHz radio, 0.96 inch SSD1306 128x64 OLED on I2C, USB-C through a CP2102 UART bridge, RST and PRG buttons, JST LiPo connector with onboard charge circuit, onboard 3.3V regulator.

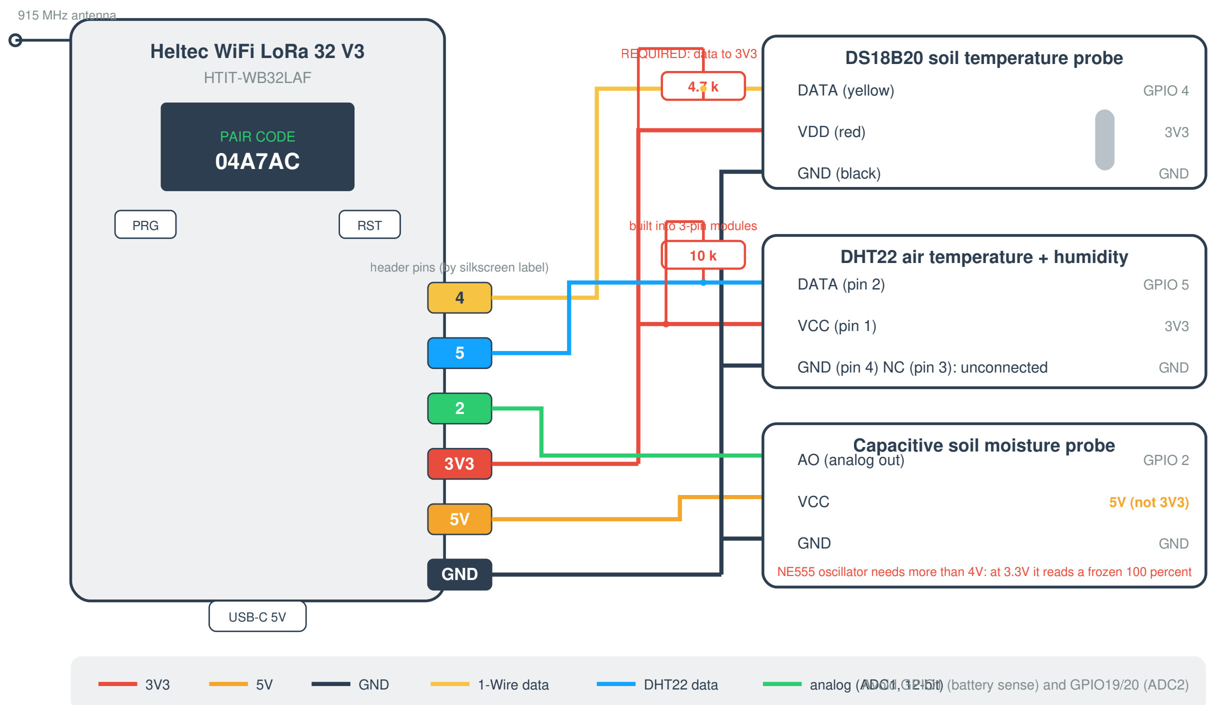
2.2 Complete GPIO map

GPIO	Function	Direction	Notes
0	PRG button	input, pull-up	Boot-strap pin; safe to read after boot; deep-sleep wake source
1	Battery sense ADC	analog in	Midpoint of the battery divider; never use for sensors
2	Soil moisture AO	analog in (ADC1)	12-bit reads, 0 to 4095
4	DS18B20 data	1-Wire	Requires external 4.7k pull-up to 3V3
5	DHT22 data	digital	Requires 10k pull-up to 3V3 (built into 3-pin modules)
7	Optional VBUS sense	analog in	Midpoint of optional 100k/100k divider from the 5V pin; read only when USE_VBUS_SENSE = 1
8 / 9 / 10 / 11	SX1262 NSS / SCK / MOSI / MISO	SPI	Fixed by board routing
12 / 13 / 14	SX1262 RST / BUSY / DIO1	control	Fixed by board routing
17 / 18 / 21	OLED SDA / SCL / RST	I2C	Fixed by board routing
36	Vext (OLED power rail)	output	Drive LOW to power the OLED
37	Battery sense control	output	Gates the battery divider; polarity inverted from the Heltec reference on this board

Constraints: analog inputs must be ADC1 channels (GPIO1 to 10); ADC2 (including GPIO19 and 20) is owned by Wi-Fi while the radio runs. ADC resolution is configured to 12 bits.

2.3 Sensor wiring specification

GrowthPulse Node Wiring (Heltec WiFi LoRa 32 V3)



Sensor	VCC	GND	Signal	Pull-up	Failure signature
DS18B20 (soil temperature)	3V3	GND	GPIO4	4.7k to 3V3, required	-127 C / -196.6 F sentinel
DHT22 (air temp + humidity)	3V3	GND	GPIO5	10k to 3V3 (built into 3-pin modules)	NaN, serialized as null
Capacitive soil moisture	**5V**	GND	GPIO2	none	raw < 300, or 10-sample spread > 600

The soil probe is powered at 5V because its NE555 oscillator requires more than about 4V; its analog output stays below about 3V, inside the ADC's safe range. DS18B20 addresses begin with family code 0x28.

2.4 Power specification

Item	Value
Input	5V via USB-C, or 5.0V at ~1A limit into the 5V/GND header pins
Draw	~150 mA idle; 300 to 500 mA bursts during Wi-Fi transmit
Sensors	Single-digit mA, from the board's regulated rails
Battery	LiPo on the JST connector; charged by the onboard circuit over USB-C
Battery voltage range	3.04V (0 percent) to 4.26V (100 percent), 100-entry discharge curve
Battery ADC conversion	volts = 16-sample raw average / 238.7
Charging detection	Inferred: rising smoothed voltage, or at/above 4.18V; pinned at/above 4.22V and flat reports AC
Deep sleep draw	Microamps (OLED rail off, wake on PRG)

3. Firmware reference (GP_Combined v4.3)

3.1 Build configuration

Setting	Value
Board	"Heltec WiFi LoRa 32(V3)" (Espressif esp32 core; boards index espressif.github.io/arduino-esp32/package_esp32_index.json)
Partition scheme	Larger-app / "Huge App" (the combined image overflows the default slot)
Upload speed	921600 default; drop to 115200 for marginal cables
Serial monitor	115200 baud
USB driver	Silicon Labs CP210x VCP

3.2 Libraries

Library	Author	Used for
WiFiManager	tzapu	Captive-portal Wi-Fi provisioning (2.0.17 at time of writing)
OneWire	Paul Stoffregen	1-Wire bus for the DS18B20
DallasTemperature	Miles Burton	DS18B20 conversion
DHT sensor library	Adafruit	DHT22 (installs Adafruit Unified Sensor)
Losant Arduino MQTT Client	Losant	Device cloud connection (installs PubSubClient)
ArduinoJson	Benoit Blanchon	Telemetry and command payloads
U8g2	oliver	OLED
RadioLib	jgromes	SX1262 LoRaWAN class-A stack

3.3 Key constants

Constant	Value	Meaning
FW_VERSION	"4.3"	Shown on the self-test and pairing screens
LORA_SUBBAND	2	US915 FSB2; must match gateway and TTS
UPLINK_FPORT	2	LoRaWAN uplink port; downlinks are read from the same exchange
LORA_UPLINK_MS	60 s	Bench/demo value; raise toward 15 min for production fair-use
WDT_TIMEOUT_S	60	Task watchdog; detached only around the captive portal
CMD_GRACE_MS	12 s	Destructive cloud commands ignored this long after connect
DIM_AFTER_MS	5 min	OLED dims to contrast 20 after idle
SELFTEST_HOLD_MS	10 s	Boot self-test screen hold
PAGE_MS / PAGE_COUNT	5 s / 3	OLED page rotation
dryValue / wetValue	3600 / 1300	Soil calibration constants (raw ADC, dry air / water)
BAT_MIN_V / BAT_MAX_V	3.04 / 4.26	Battery curve endpoints
USE_VBUS_SENSE	0	Set to 1 only on boards with the 5V-to-GPIO7 divider
Wi-Fi telemetry interval	3 s	Loop cadence in Wi-Fi mode
Portal timeout	600 s	Setup hotspot window
Join retry backoff	10 s	Between activateOTAA attempts
Provisioning retry	5 s	Between provision-device attempts

3.4 NVS storage map

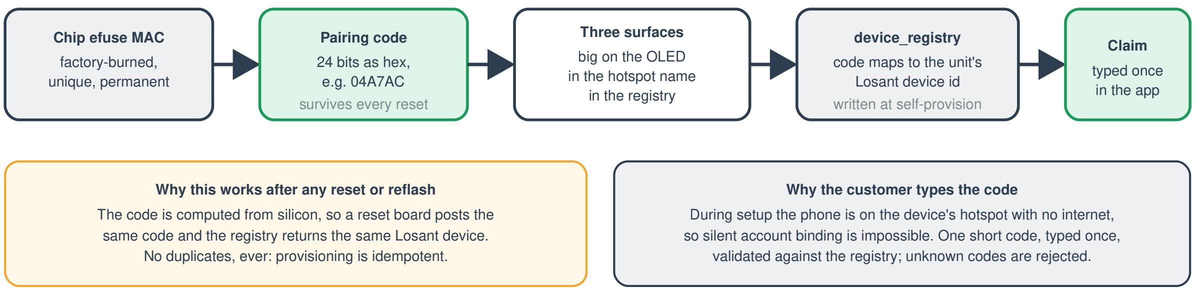
Namespace	Key	Contents
gp	netmode	"wifi" (default) or "lorawan"
gp	ldid / lkey / lsec	Losant device id, access key, access secret (from self-provisioning)
gp	lwJoinEUI / lwDevEUI / lwAppKey	OTAA keys (pushed by provisionLoRa)
lorawan	nonces	RadioLib LoRaWAN nonce buffer (DevNonce persistence)
(WiFiManager)		Saved Wi-Fi SSID and password, managed by the library

Wi-Fi credentials, the Losant identity, and the nonces all survive reflashes; a full NVS erase (or factory reset) clears them.

3.5 Identity derivation

`pairCode()` = uppercase hex, zero-padded to 6 characters, of bits 24 to 47 of the efuse MAC (`(ESP.getEfuseMac() >> 24) & 0xFFFFFF`). The setup hotspot is `GrowthPulse-<pairCode>`. The code is permanent and survives every reset; the registry maps it to the Losant device id. Legacy units provisioned under the original firmware keep their registry codes (the demo unit's is the five-character 4A7AC).

Identity and Claiming (one permanent identity, three surfaces)

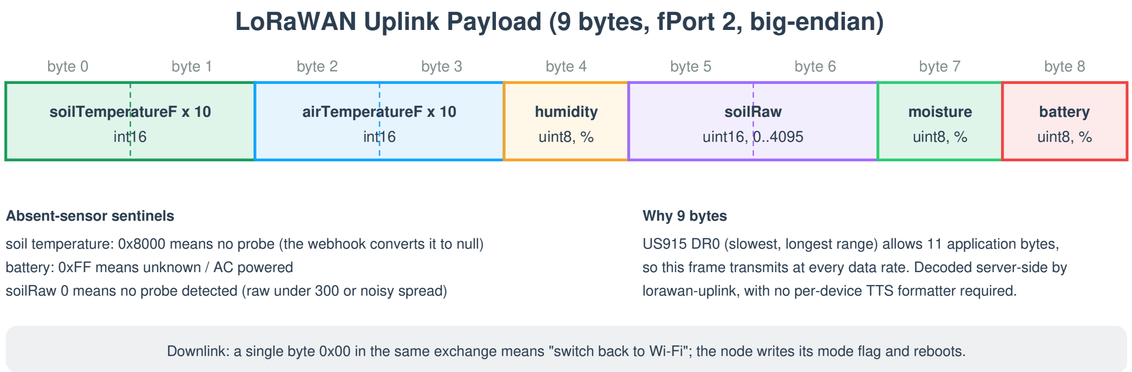


3.6 Wi-Fi telemetry schema (MQTT state, every 3 s)

```
{
  "soilTemperatureF": 72.5,
  "airTemperatureF": 78.8,
  "airHumidity": 52.0,
  "soilRaw": 2410,
  "soilMoisturePercent": 42,
  "wifiRssi": -55,
  "batteryPct": 87,
  "charging": false,
  "transport": "wifi"
}
```

Sentinels pass through unmodified: soilTemperatureF is -196.6 when the DS18B20 is absent; airTemperatureF/airHumidity are null when the DHT22 is absent; soilRaw is 0 when the probe is absent (raw < 300 or spread > 600); batteryPct is -1 internally and reported as AC when no battery is detected.

3.7 LoRaWAN uplink payload (9 bytes, fPort 2)



Bytes	Field	Encoding	Absent value
0..1	soilTemperatureF x 10	int16 big-endian	0x8000
2..3	airTemperatureF x 10	int16 big-endian	0
4	airHumidity percent	uint8	0
5..6	soilRaw	uint16 big-endian (0..4095)	0
7	soilMoisturePercent	uint8 (0..100)	0
8	batteryPct	uint8 (0..100)	0xFF

Nine bytes fits the 11-byte DR0 limit, so the frame transmits at every US915 data rate. The webhook decodes these bytes itself; no TTS payload formatter is required.

3.8 Downlink specification

Payload	Port	Effect
One byte, 0x00	fPort 2 (read from the uplink exchange)	Write netmode = wifi and reboot

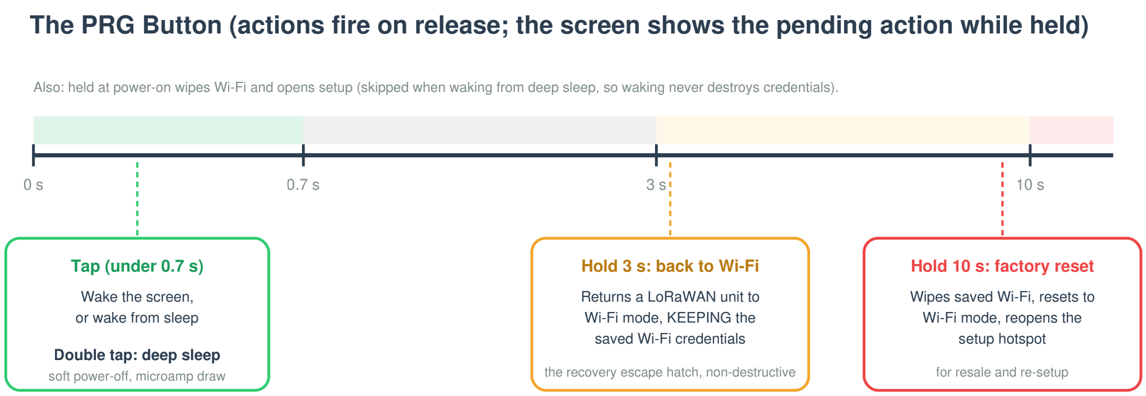
Downlinks are queued through the TTS Application Server `down/replace` endpoint and delivered in the class-A receive window after the node's next uplink. Latency is bounded by LORA_UPLINK_MS.

3.9 Cloud command specification (Losant, Wi-Fi mode only)

Command	Payload	Effect
factoryReset	none	Wipe Wi-Fi credentials, set netmode = wifi, reboot into setup
setMode	`{ "mode": "wifi" | "lorawan" }`	Write netmode and reboot
provisionLoRa	`{ "joinEUI": "16 hex", "devEUI": "16 hex", "appKey": "32 hex" }`	Validate, store keys in NVS, reboot into LoRaWAN

All three are ignored during the first 12 seconds after connecting to Losant (stale-replay grace window). Unknown commands are ignored.

3.10 PRG button specification



Gesture	Threshold	Action
Single tap	< 700 ms	Wake screen / wake from deep sleep
Double tap	two taps within 600 ms	Enter deep sleep (soft power-off)
Hold	3 s to 10 s	Switch to Wi-Fi mode, keep saved Wi-Fi credentials
Hold	10 s or more	Factory reset: wipe Wi-Fi, netmode = wifi, reboot to setup
Held at power-on		Wipe Wi-Fi and open setup (skipped when waking from deep sleep)

Actions fire on release; the OLED shows the pending action while the button is held.

3.11 OLED screens

Rotating pages (5 s each): pair code (with link chip and firmware version), connection (link, signal, battery/AC), live readings. Event screens: starting up / waking up, self-test (10 s, OK or "--" per sensor), setup instructions with the hotspot name, connecting, provisioning, switching messages, button hints, sleeping. Display dims after 5 idle minutes; any tap wakes it.

3.12 Serial output reference

115200 baud. Notable lines: the boot banner with mode, `Wi-Fi IP: ...`, `First-time setup / registering device`, `Provisioned. Losant device id: ...`, `Cloud command: <name>` (with "ignored: stale command" when inside the grace window), `LoRaWAN: OTAA join (US915 FSB2)...`, `LoRaWAN: JOINED.`, `Uplink sent; downlink RX1/RX2. RSSI=.. SNR=..`, `Join failed (<code>). Retry in 10s.`, and the per-cycle telemetry summary used for soil calibration.

3.13 LoRaWAN radio configuration

Item	Value
Stack	RadioLib LoRaWANNode, class A, OTAA
Region / sub-band	US915 / FSB2 (uplink channels 8 to 15)
MAC / PHY version	LoRaWAN 1.0.4 / RP002 1.0.4 (nwkKey = appKey for 1.0.x)
JoinEUI	0000000000000000
TCXO	RadioLib default 1.6V (required non-zero on the V3)
RF switch	setDio2AsRfSwitch(true)
Duty cycle	setDutyCycle(false); US915 is governed by TTN fair use (30 s airtime/day, 10 downlinks/day)
Nonce persistence	NVS, saved after every join attempt and uplink

4. Cloud API reference (Netlify functions)

All endpoints are same-origin under `https://growthpulsecloud.com/.netlify/functions/`. "Session" means the caller's Supabase JWT in an `Authorization: Bearer` header, validated against `auth/v1/user`.

4.1 device-state

Method	GET `?deviceId=<losantDeviceId>`
Auth	none (read-only token server-side)
Returns	`{ airTemperatureF, airHumidity, soilTemperatureF, soilRaw, soilMoisturePercent, batteryPct, charging, wifiRssi, loraRssi, loraSnr, transport, time }`, missing/mistyped values as null, `Cache-Control: no-store`
Errors	400 missing deviceId, 500 unconfigured, upstream status passthrough

4.2 device-command

Method	POST `?deviceId=<losantDeviceId>&name=<command>`
Auth	Session required; ownership verified by querying `devices` with the caller's JWT (RLS decides)
Allowlist	factoryReset
Errors	401 no session, 403 not the owner, 400 unknown command, 405 non-POST

4.3 device-history

Method	GET `?deviceId&start&end` (ms epoch)
Auth	none (read-only token server-side)
Behavior	Losant time-series query, MEAN aggregation, resolution auto-scaled to <= ~240 points (minimum 1-minute buckets); sentinels cleaned to null; empty lead/tail buckets trimmed; interior gaps preserved

4.4 render-pdf

Method	POST `{ html, filename }`
Auth	Session required (401 otherwise)
Behavior	Headless Chromium (@sparticuz/chromium + puppeteer-core, shipped via external_node_modules), letter format, printBackground, returns base64 PDF as attachment

4.5 provision-device

Method	POST `{ "code": "<pairing code>", "token": "<shared firmware token>" }`
Auth	The shared PROVISION_TOKEN
Behavior	Look up code in device_registry (service role); create Losant device "GrowthPulse CODE" with the full attribute schema if new and register the mapping; PATCH the attribute schema on every call (self-heal); mint a device-scoped access key
Returns	`{ deviceId, accessKey, accessSecret }`, no-store
Errors	401 bad token, 400 bad/missing code, 502 with detail (Losant create, registry insert, key create)

4.6 provision-lorawan

Method	POST `{ "losantDeviceId": "<24 hex>" }`
Auth	Session required
Reuse path	Route found in lorawan_devices with stored app_key: clear the TTS downlink queue (down/replace with []), PUT resets_join_nonces on the **Join Server**, re-push stored keys via the Losant provisionLoRa command; returns `{ ok, ttsDeviceId, devEUI, reused: true }`
Create path	Generate DevEUI (8 random bytes) and AppKey (16 random bytes), JoinEUI zeros, TTS device id `gp-<deveui lowercase>`; four-call create (IS on eu1, then JS, NS, AS on the cluster); upsert the route (tts_device_id, dev_eui, app_key, losant_device_id, user_id); push keys via provisionLoRa
Errors	401, 400, 500 unconfigured, 502 with the failing TTS call's detail; a failed route store is surfaced (never swallowed); the AppKey is never returned to the browser

4.7 lorawan-uplink (webhook target)

Method	POST (called by TTS)
Auth	X-Webhook-Token header must equal LORAWAN_WEBHOOK_TOKEN
Behavior	Ignore non-uplink events; use decoded_payload if present, else decode the 9 bytes from frm_payload; resolve the Losant device via LORAWAN_DEVICE_MAP (static), then lorawan_devices (service role), then LORAWAN_DEFAULT_DEVICE_ID; clean sentinels; build state tagged transport=lorawan with loraRssi/loraSnr from rx_metadata[0]; omit all null fields; POST to Losant state; log the outcome per delivery
Errors	401 bad token, 422 undecodable, 404 no mapping, 502 Losant rejection (with detail)

4.8 lorawan-switch-wifi

Method	POST `{ "losantDeviceId": "<24 hex>" }`
Auth	Session required
Behavior	Look up the TTS device in lorawan_devices; POST down/replace with one downlink `{ f_port: 2, frm_payload: "AA==", priority: NORMAL }`
Errors	404 no route ("was it provisioned?"), 502 TTS rejection

4.9 register-gateway

Method	POST `{ "eui": "<16 hex>", "name": "<display name>" }`
Auth	Session required
Behavior	Normalize the EUI; create gateway `eui-<eui lowercase>` via the Identity Server under TTS_USER_ID with the cluster as gateway server, frequency plan TTS_FREQ_PLAN, unauthenticated connections allowed (Semtech UDP); 409 already-registered returns success

4.10 Environment variables

Variable	Scope	Purpose
VITE_SUPABASE_URL / VITE_SUPABASE_ANON_KEY	public (bundle)	Supabase project and anon key (RLS-gated)
SUPABASE_SERVICE_ROLE_KEY	server	Registry/route/ownership access bypassing RLS
LOSANT_APP_ID	server	Losant application id
LOSANT_API_TOKEN	server	Read-only Losant token (state, history)
LOSANT_COMMAND_TOKEN	server	Write-capable Losant token (commands, webhook state)
LOSANT_PROVISION_API_TOKEN	server	Losant token with devices. *and applicationKeys.*
PROVISION_TOKEN	server	Shared firmware provisioning secret
TTS_API_KEY	server	TTS key with end-device and gateway write rights
TTS_APP_ID	server	growthpulse
TTS_CLUSTER	server	nam1.cloud.thethings.network (default)
TTS_IS_HOST	server	eu1.cloud.thethings.network (default)
TTS_USER_ID	server	TTS account owning registered gateways
TTS_FREQ_PLAN	server	US_902_928_FSB_2 (default)
LORAWAN_WEBHOOK_TOKEN	server	Webhook shared secret
LORAWAN_DEVICE_MAP / LORAWAN_DEFAULT_DEVICE_ID	server, optional	Static webhook routing fallbacks

Environment changes require a redeploy to take effect.

5. Database schemas (Supabase)

5.1 device_registry

```
create table device_registry (
  claim_code        text primary key,
  losant_device_id  text not null,
  created_at        timestamptz default now()
);
-- RLS: authenticated may SELECT; no client write policies.
```

5.2 devices

```
create table devices (
  id                uuid primary key default gen_random_uuid(),
  user_id           uuid not null default auth.uid() references auth.users(id) on delete
cascade,
  claim_code        text not null unique,
  losant_device_id  text not null,
  name              text not null default 'My Plant',
  location          text default '',
  geo               jsonb,
  grp               text,
  transport         text not null default 'wifi',
  plant             text not null default 'generic',
  irrigation        jsonb,
  photo             text,
  created_at        timestamptz default now()
);
-- RLS: per-user policies on select / insert / update / delete (auth.uid() = user_id).
```

5.3 gateways

```
create table gateways (
  id          uuid primary key default gen_random_uuid(),
  user_id     uuid not null default auth.uid() references auth.users(id) on delete cascade,
  name        text not null default 'My Gateway',
  eui         text not null,
  created_at  timestamptz default now()
);
-- RLS: per-user policies on all four verbs.
```

5.4 lorawan_devices (v2)

```
create table lorawan_devices (
  tts_device_id     text primary key,
  dev_eui           text not null,
  app_key           text,
  losant_device_id  text not null,
  user_id           uuid references auth.users(id) on delete cascade,
  created_at        timestamptz default now()
);
create unique index lorawan_devices_losant_uidx on lorawan_devices (losant_device_id);
-- RLS enabled with no client policies: service-role (functions) only.
```

User metadata on `auth.users`: `first_name`, `last_name`, `grower_type` (hobbyist / home grower / farmer / commercial), `tier` (free / plus / pro).

---

6. Losant reference

Item	Value
Device attributes	airTemperatureF, airHumidity, soilTemperatureF, soilRaw, soilMoisturePercent (number); wifiRssi, loraRssi, loraSnr, batteryPct (number); charging (boolean); transport (string)
Telemetry topic	losant/{deviceId}/state (MQTT, device access key)
Command topic	losant/{deviceId}/command (MQTT; no offline queue)
State read	GET /applications/{appId}/devices/{deviceId}/compositeState (read-only token)
State write	POST /applications/{appId}/devices/{deviceId}/state (write token; used by the webhook)
Command send	POST /applications/{appId}/devices/{deviceId}/command (write token)
History	POST /applications/{appId}/data/time-series-query (read-only token)
Device create / key create	POST /applications/{appId}/devices and /applications/{appId}/keys (provisioning token)
Validation rule	A state report containing an undefined attribute is rejected whole; a null for a typed Number attribute is rejected
Email alert rate limit	One email per minute per workflow

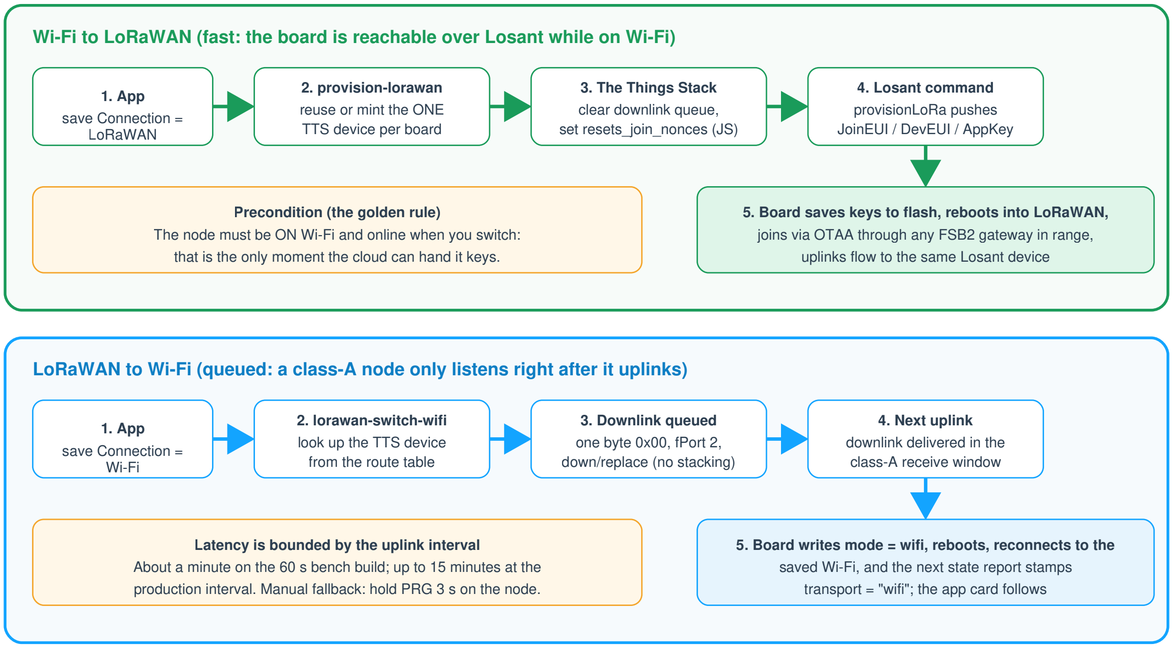
Alert workflows: Device State trigger, conditional (for example `{{ data.soilMoisturePercent }} < 25`), email/SMS node on the true branch.

7. The Things Stack reference

Item	Value
Identity Server (registrations)	eu1.cloud.thethings.network
Operating cluster (NS/JS/AS, traffic)	nam1.cloud.thethings.network
Console	https://nam1.cloud.thethings.network/console
Frequency plan	US_902_928_FSB_2 (uplink channels 8 to 15 plus 500 kHz channel 65)
Device settings	OTAA, MAC 1.0.4, RP002 1.0.4, supports_join, resets_join_nonces = true (a Join Server field)
Device id convention	gp- (auto-provisioned); gateway id eui-
Gateway forwarder	Semtech UDP packet forwarder, server nam1.cloud.thethings.network, ports 1700/1700
Webhook	JSON, base URL https://growthpulsecloud.com/.netlify/functions/lorawan-uplink, uplink message enabled, header X-Webhook-Token
Webhook health	TTS auto-deactivates a webhook after repeated delivery failures (circuit breaker); check the status banner
Fair use (TTN sandbox)	~30 s uplink airtime per device per day; 10 downlinks per device per day
US915 payload limits	DR0 = 11 bytes; DR1 = 53; DR2 = 125; DR3/DR4 = 222
Downlink queue ops	POST .../as/applications/{app}/devices/{id}/down/replace (clear with `{"downlinks":[]}`)

The four-call device creation sequence and field-mask ownership rules are specified in section 4.6 and implemented in `provision-lorawan.js`.

Transport Switching, Both Directions



The plant stays the same plant in the app throughout: both transports write to the same Losant device.

8. Web application reference

8.1 Stack and structure

React 18, Vite 5, recharts, lucide-react, @supabase/supabase-js, html2pdf.js (dynamic import). No router; a five-tab shell (Live, History, Alarms, Devices, Settings). Entry `src/main.jsx` and `src/App.jsx` (Gate + Shell); store in `src/store/` (AppContext, helpers, plants, tiers); auth in `src/auth/AuthProvider.jsx`; views in `src/views/`; components in `src/components/`; utilities in `src/utils/`.

8.2 Behavioral constants

Item	Value
Default refresh interval	2 s (user options 1 s to 5 min; fast rates disabled without a Wi-Fi device)
Offline threshold	45 s (Wi-Fi), 15 min (LoRaWAN)
Connecting grace window	15 s after load for newly claimed devices
In-memory history ring	60 readings per device
Health score weights	good 100, warn 66, critical 28, connected metrics only
Battery warnings	warn at 20 percent, critical at 10, suppressed while charging/AC
Rain-delay threshold	Rain chance >= 50 percent triggers the pause prompt
Soil ADC mock constants	DRY 3600, WET 1300 (mirrors firmware)
Report cap	~240 chart points per series
Journal photo thumbnails	Max 720 px, JPEG quality 0.7 (fileToThumb)
Toast duration	4 s, keyed by deviceId:ruleId

8.3 localStorage keys

All prefixed `growthpulse:<userId>:` for account isolation (demo uses its own synthetic user id): devices (demo only), selectedDeviceId, alarmRules, settings, tier (cache; Supabase metadata is the source of truth), journals, gateways (demo only).

8.4 Subscription tiers

Tier	Price	Devices	Unlocks
Free	$0	3	Plant catalog, live dashboard, alarms, history
Plus	$4.99/mo	10	Weather rain gauge, rain and sun alerts
Pro	$9.99/mo	99	Automated irrigation with rain delay, LoRaWAN gateways

Demo mode runs Pro. Billing is a roadmap item; tier selection persists in user metadata.

8.5 Plant catalog ranges

Ten categories; each species maps to one. Ranges are (good band, warn band):

Category	Soil moisture %	Humidity %	Air temp F
Houseplants	40-70, 30-80	40-65, 30-78	65-80, 58-88
Tropical and ferns	50-80, 40-90	55-85, 45-92	65-82, 58-90
Succulents and cacti	5-30, 3-45	20-45, 10-55	65-90, 55-100
Herbs	40-70, 30-80	40-65, 30-75	65-82, 55-90
Vegetables	45-75, 35-85	50-75, 40-85	60-85, 50-92
Fruits and berries	45-75, 35-85	45-70, 35-80	60-85, 45-92
Flowers	40-70, 30-80	40-70, 30-80	60-82, 50-90
Orchids	40-65, 30-75	55-80, 45-90	65-82, 58-90
Trees and shrubs	35-65, 25-78	35-70, 25-82	50-88, 35-95
Lawn and grass	40-70, 30-82	35-75, 25-85	55-88, 45-95

Auto-set alarms derive from the plant's good band: soil below good-low, air temperature above good-high, humidity below good-low.

8.6 Disconnected-sensor signatures (metricConnected)

Metric	Disconnected when
Soil temperature	value < -100 (the -127 C / -196.6 F sentinel)
Air temperature / humidity	null, or humidity <= 0
Soil moisture	soilRaw < 300

8.7 Third-party APIs (no keys)

API	Endpoint
Open-Meteo forecast	GET api.open-meteo.com/v1/forecast?latitude&longitude&current=...&daily=...&temperature_unit&timezone=auto&forecast_days=1
Open-Meteo geocoder	GET geocoding-api.open-meteo.com/v1/search?name=&count=1&language=en

Weather is fetched for the selected device's stored geo only; the app never reads browser geolocation. WMO weather codes map to label, icon, and color in WeatherCard.

8.8 PWA

manifest.webmanifest (standalone display, theme/background colors, 192 and 512 maskable icons), apple-touch-icon at 180 px on a white matte, safe-area insets throughout, version stamp (version + build date) in Settings.

---

9. Glossary

Term	Meaning
Pairing code / claim code	The chip-derived uppercase hex code on the OLED; the unit's public identity and serial
Self-provisioning	A board minting its own cloud identity at first boot via provision-device
Transport	The link a node currently uses: wifi or lorawan; stamped on every state report
OTAA	Over-the-air activation; the LoRaWAN join handshake that derives session keys
DevEUI / JoinEUI / AppKey	The OTAA identity triplet; AppKey lives on the Join Server and in board NVS only
DevNonce	The monotonically increasing join counter; persisted in NVS, tolerated across resets via resets_join_nonces
FSB2	US915 frequency sub-band 2, the TTN channel plan (uplink channels 8 to 15)
Class A	The LoRaWAN device class: receive windows exist only after an uplink
Sentinel	A measured failure signature passed through the pipeline (for example -196.6 F)
Route	A lorawan_devices row mapping a TTS device to its Losant device
Composite state	Losant's latest-value-per-attribute view of a device
RLS	Supabase row-level security; the authorization layer for every table